

### 3. Quadratic Limb Darkening

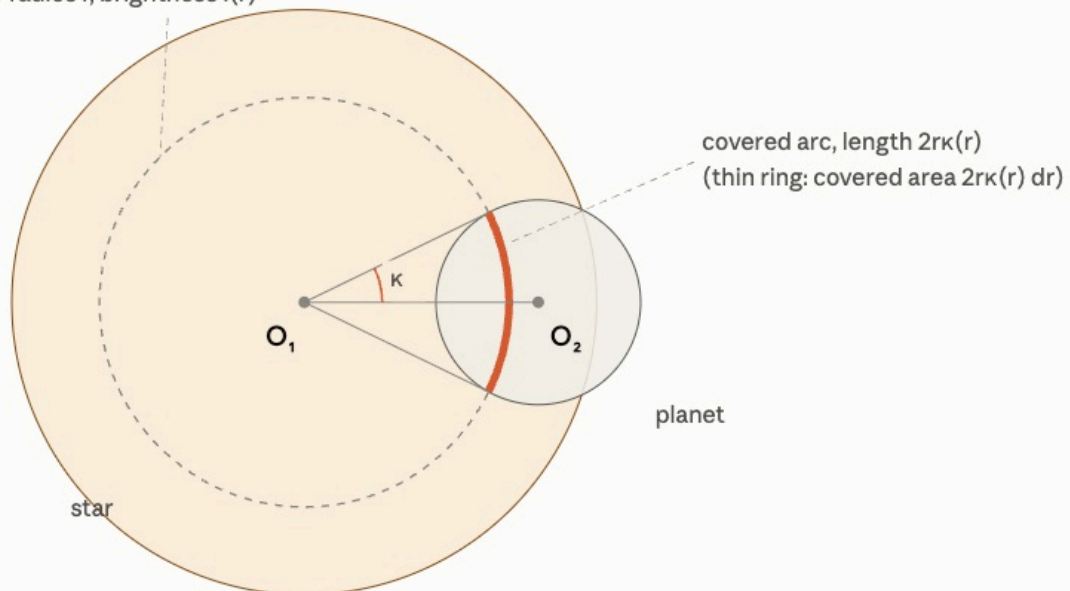
This model is much more realistic. Stars appear darker towards their edge (this is limb darkening), because near the limb our line of sight only reaches cooler outer layers of the star. Here, I describe the brightness across the disk with the standard quadratic law,

$$I(r) = 1 - g_1(1-\mu) - g_2(1-\mu)^2.$$

This is the simplest curve that equals 1 at the centre and has enough freedom to fit real stars. The coefficients  $g_1$  and  $g_2$  will be taken from published tables.

Setting:

ring of radius  $r$ , brightness  $I(r)$



# Brightness or Intensity $I(r)$ is

$$I_\nu = \frac{dE}{dt \cdot dA \cos \theta \cdot d\Omega \cdot d\nu}$$

Where:

- $dE$  is the **energy**
- $dt$  is the **unit time**
- $dA \cos \theta$  is the **projected area** perpendicular to the ray (where  $\theta$  is the angle between the surface normal and the ray)
- $d\Omega$  is the **unit solid angle** (the direction of the ray)
- $d\nu$  is the **unit frequency interval**

## Assumption:

For limb-darkening, the brightness at the same radius  $r$  from the centre is equal.

So we can use the concept of a ring to compute overall brightness when eclipse occurs.

We have:

$$F(p, z) = 1 - \frac{\text{blocked\_brightness}}{\text{total\_brightness}}$$

We will use an integral where the radius  $r$  slides from 0 to 1 and we calculate the brightness blocked from an overlap on the ring at radius  $r$  from centre and width  $dr$ .

Therefore:

$$\text{total\_brightness} = \int_0^1 I(r) \cdot 2\pi r \, dr$$

(since the total brightness is for full ring, length  $2\pi r$ )

Now, let  $k(r)$  be the function returning the half-angle of the ring fraction that holds an overlap at radius  $r$ . Then

$$\begin{aligned} \text{blocked-brightness} &= \int_0^1 I(r) \cdot 2\pi r \cdot \frac{2 \cdot k(r)}{2\pi} dr \\ &= \int_0^1 I(r) \cdot 2\pi r k(r) dr \end{aligned}$$

Geometrically, from cosine rule we can get:

$$k(r) = \begin{cases} \pi & r \leq p - z \\ \cos^{-1}\left(\frac{r^2 + z^2 - p^2}{2rz}\right) & |z - p| < r < z + p \\ 0 & \text{otherwise} \end{cases}$$

Hence we get our final relative flux formula:

$$F(p, z) = 1 - \frac{\int_0^1 I(r) \cdot 2\pi r k(r) dr}{\int_0^1 I(r) \cdot 2\pi r dr}$$